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Software Test Plan

Project: ResiTrack — Apartment / Residencies Organizer **Format:** IEEE 829-1998 **Course:** Software Engineering

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1. Test Plan Identifier

`RESITRACK-TP-001` — covers all requirements in `SRS.docx` (FR-01–FR-15, NFR-01–NFR-10).

2. Introduction

Defines the scope, approach, and deliverables for testing ResiTrack, a MERN application serving Administrator, Resident, Maintenance Staff, and Owner roles. Security requirements are validated as a dedicated activity (§5.1), not just a byproduct of functional testing.

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3. Test Items

Item	Covers
Authentication & RBAC	FR-02, FR-14, NFR-03–05, SEC-01–04
Tenant & Flat Management	FR-01, FR-12
Notices	FR-03
Maintenance	FR-04, FR-05, FR-09
Rent & Receipts	FR-06, FR-07, FR-15
Visitor Log	FR-08
Complaints	FR-13
Owner Reporting	FR-10
Documents	FR-11, SEC-06
REST API layer	All endpoints backing the above

4. Features to be Tested / Not Tested

In scope: all FR-01–FR-15, NFR-01–NFR-10, and SEC-01–SEC-06 (SRS §5.2).

Out of scope this cycle: multi-complex/multi-tenancy, online payment gateway integration (payments are recorded manually, not collected in-app), native mobile apps, third-party email/SMS provider internals (only the integration trigger is tested).

5. Approach

Layered strategy: **unit tests** (controllers, ≥60% coverage per NFR-09) → **API/integration tests** (endpoints + RBAC middleware against a test DB) → **system/E2E tests** (UI flows per use case) → **non-functional tests** (load, availability, usability, responsiveness) → **acceptance walkthrough**.

Entry criteria: feature merged, unit tests passing in CI. *Exit criteria:* all planned test cases executed; all High-priority cases pass; no open Critical/High-severity defects.

5.1 Security Validation

Control	Validation Method	Requirement
Password storage	Inspect DB; confirm bcrypt hash prefix `$2b$12$`, never plaintext.	SEC-01
Transport encryption	TLS scan (SSL Labs / `testssl.sh`); confirm TLS 1.2+ only, HTTP rejected/redirected.	SEC-02
RBAC enforcement	Call every other role's protected endpoints directly (Postman); confirm HTTP 403.	SEC-03
Session timeout	Idle 30+ minutes, attempt an action; confirm token rejected, redirect to login.	SEC-04
Audit logging	Perform admin actions; confirm immutable, retrievable log entries with actor ID + timestamp.	SEC-05
File-upload validation	Attempt disallowed file types / >5 MB uploads; confirm rejection, nothing persisted.	SEC-06

Run once during the normal test cycle and again as a dedicated pass before final submission.

6. Plan Elements (Summary)

Element	Summary
Pass/Fail Criteria	A case passes only if actual results match expected results exactly, including error paths.

Element	Summary
Suspension/Resumption	Suspend if a Critical defect blocks ≥30% of remaining cases, the environment is unavailable, or a security control fails; resume once fixed/restored.
Deliverables	This plan, test case log, defect log, security validation report, final summary report.
Environment	Staging deployment separate from any demo instance; Node.js, MongoDB test cluster, Postman, a load-test tool (k6/Artillery), an SSL scanner; synthetic (non-PII) test data.
Responsibilities	Shared across the 4-team members, with peer review of test cases before execution.
Schedule	Test authoring alongside development (Weeks 5–8); execution (Weeks 9–13); dedicated security + regression pass (Week 14); final report by Week 15, ahead of Week 16 submission.
Risks	Free-tier hosting limits load-test realism (mitigate: test locally with equivalent caps); limited tooling budget for security scans (mitigate: use free tools — `testssl.sh`, OWASP ZAP baseline).
Approvals	Sign-off by all four team members before final submission.

7. Test Cases

Traceable directly to the SRS (Req ID + Use Case columns = traceability; see also SRS §6 RTM).

TC ID	Req ID	Use Case	Description	Expected Result	Priority
TC-01	FR-01	UC-01	Register a new tenant	Record created in DB with correct fields; success message shown	High
TC-02	FR-02	UC-02	Login with valid/invalid credentials	Valid login succeeds; invalid login rejected, no	High

TC ID	Req ID	Use Case	Description	Expected Result	Priority
				session created	
TC-03	FR-03	UC-03	Post a community notice	Notice visible to all residents within 5s	High
TC-04	FR-04	UC-04	Raise a maintenance request	Request saved as “Open”; confirmation shown	High
TC-05	FR-05	UC-05	Update maintenance request status	Status change reflected in DB and resident view within 3s	High
TC-06	FR-06	UC-06	Record a rent payment	Payment record correct; tenant marked paid	High
TC-07	FR-07	UC-07	Automated payment reminder	Reminder triggered ≥3 days before due date	Medium
TC-08	FR-08	UC-08	Log a visitor entry	Entry appears in admin dashboard with timestamp	Medium
TC-09	FR-12	UC-12	Manage flat inventory	Add/edit/ deactivate reflected correctly; deactivated flat not assignable	High
TC-10	FR-14	UC-14	Role-based access control	Out-of-role API call rejected with	High

TC ID	Req ID	Use Case	Description	Expected Result	Priority
				HTTP 403	
TC-NFR-01	NFR-01	—	Response time under load	95th-percentile response ≤3s at 200 concurrent users	High
TC-NFR-03	NFR-03 / SEC-01	UC-02	Password storage security	Password field is a bcrypt hash (cost ≥12), never plaintext	High
TC-NFR-04	NFR-04 / SEC-02	—	Transport encryption	TLS 1.2+ confirmed; plain HTTP rejected/redirected	High
TC-NFR-05	NFR-05 / SEC-04	UC-02	Session timeout	Idle 30+ min session rejected; redirected to login	High
TC-NFR-06	NFR-06	All UCs	Mobile responsiveness	Key screens usable at 375px width, no clipped controls	Medium